IoT and Digitalization for the Circular Economy

Lecture 0: Organization

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M.Sc. Anant Sujatanagarjuna
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- Updated versions of these slides will be available in our Github repository.

Team

Prof. Dr. Benjamin Leiding

M.Sc. Anant Sujatanagarjuna

M.Sc. Chintan Patel

Research Group

- **E**merging **T**echnologies for the **C**ircular **E**conomy → **ETCE**
- Research focus:
  - Intersection of IT and sustainability
  - Circular Economy and Circular Societies
  - Self-organized, decentralized and distributed systems
  - Sustainable and resilient food production
- Other courses:
  - Requirements Engineering (WS – M.Sc.)
  - The Limits to Growth – Sustainability and the Circular Economy (WS – open for everyone)

Research Group

- ETCE Website – Link
  - Course material
  - Theses/project topics

Learning Outcome

- Understanding the concept of a circular economy, sustainability, and related concepts (biocapacity, etc.).
- Gain a basic understanding of causes, dimensions, and the characterization of climate change, environmental pollution, and dwindling non-renewable resources.
- Introduction to IoT and cyberphysical systems in the circular economy
- Sensors and actuators for IoT, control and process systems of the circular economy
- Experience in prototyping IoT applications and systems
- The ability to critically assess upcoming technological solutions enabling/facilitating sustainability and the circular economy.

Lectures

- 27.04.2026 → Organization (L00)
- 04.05.2026 → Introduction (L01)
- 11.05.2026 → Circular Economy (L02)
- 18.05.2026 → Lifecycle Assessment – LCA I (L03)
- 01.06.2026 → Lifecycle Assessment – LCA II (L04)
- 08.06.2026 → Introduction to the Internet of Things (L05)
- 15.06.2026 → Internet of Things – Communication (L06) + Security and Privacy (L07)
- 22.06.2026 → Internet of Things – Data Processing and Big Data (L08)
- **29.06.2026 → IoT in Mining I (L09)**
- **06.07.2026 → IoT in Mining II (L10)**
- 13.07.2026 → Digital Technologies and Greenwashing (L11)
- **23.07.2026 → Coding Workshop I (Goslar)**
- **24.07.2026 → Coding Workshop II (Goslar)**
- 27.07.2026 → Exam Q&A

Exercises (Deadlines)

- 18.05.2026 → Exercise 01 – Circular Economy (MC)
- 01.06.2026 → Exercise 02 – Lifecycle Assessment (LCA) – Part 1
- 08.06.2026 → Exercise 03 – Lifecycle Assessment (LCA) – Part 2
- 13.07.2026 → Exercise 04 – IoT in Mining
- 20.07.2026 → Exercise 05 – TBA

Course Organization

- News and updates → StudIP

- Organization of the lecture:
  - Slides are available on Github (Link)
    - Please report bugs!
  - Lectures and exercises as live stream (BBB – next slide)
  - Lecture recordings will be available on StudIP and on Github
  - Exercise time slots = Time for questions and eventual tutorials related to the exercises

Questions? Write us an email: etce-iot@tu-clausthal.de ← **We will only respond to emails written to this specific email address!**

Dates/Times/Locations

- Lecture:
  - Monday **2:15 pm to 3:45 pm** (Berlin time) – **27.04.2026** to **13.07.2026**
  - Location: Goslar Gotec (Am Stollen 19 C, 38640 Goslar, Germany) or via BigBlueButton (Link) **OR video recordings**
- Exercise / Q&A:
  - Monday **4 pm to 4:45 pm** (Berlin time) – **27.04.2026** to **13.07.2026**
  - Location: Goslar Gotec (Am Stollen 19 C, 38640 Goslar, Germany) or via BigBlueButton (Link)
- Practical Coding Workshop:
  - When: **23.07.2026 and 24.07.2026** from **10 am – 5 pm (Berlin time)**
  - Location: Goslar (DIGIT – Wallstr. 6, 38640 Goslar)

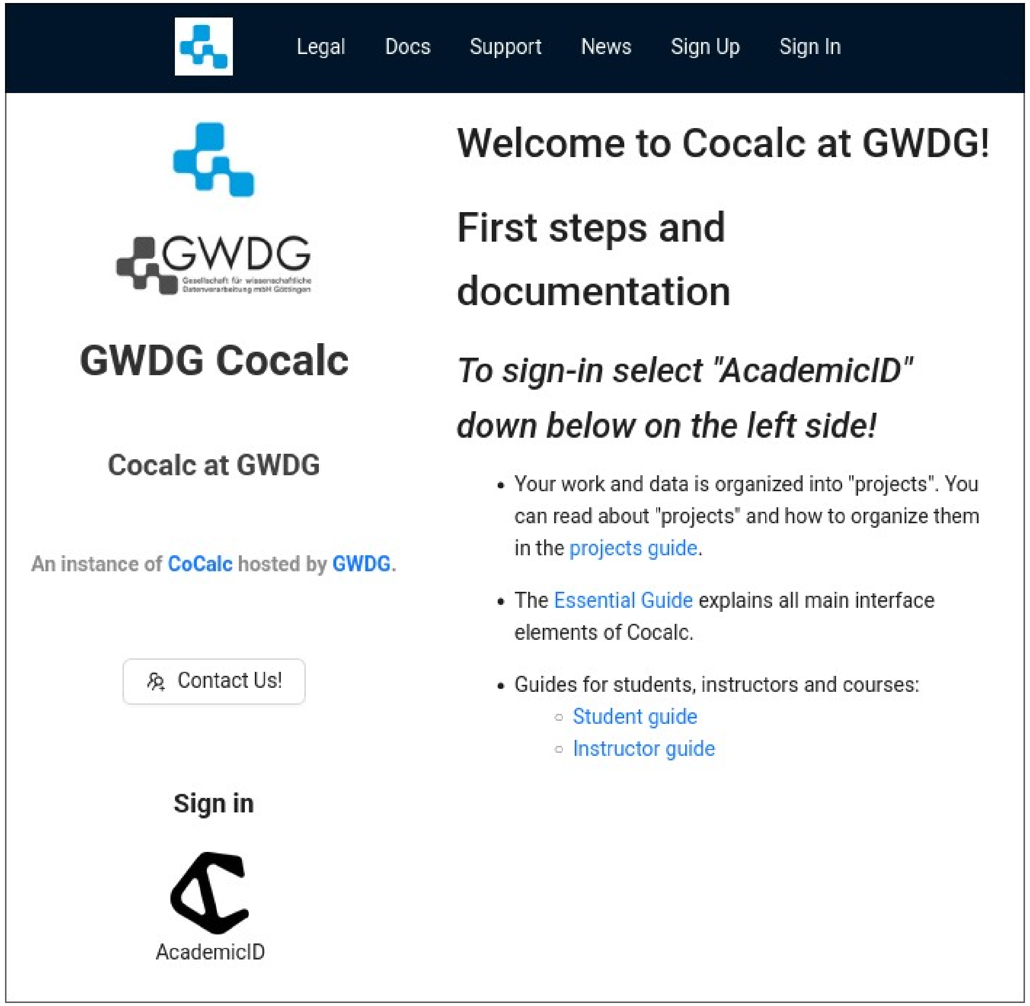
Exercises and Practical Workshop

- Individual work → no group submissions
- Multiple-Choice or written exercises
- Multiple-Choice exercises are considered “self-evaluations” and are not submitted
- Submission of **each** written exercise is **mandatory**
- **You pass by receiving a grade greater than 0.**
- You will receive feedback on your submission (during Q&A session)
- Exercise = learning feedback
- Practical workshop → You pass the workshop if you score 50% (or more)

Exercise Submission and Grading

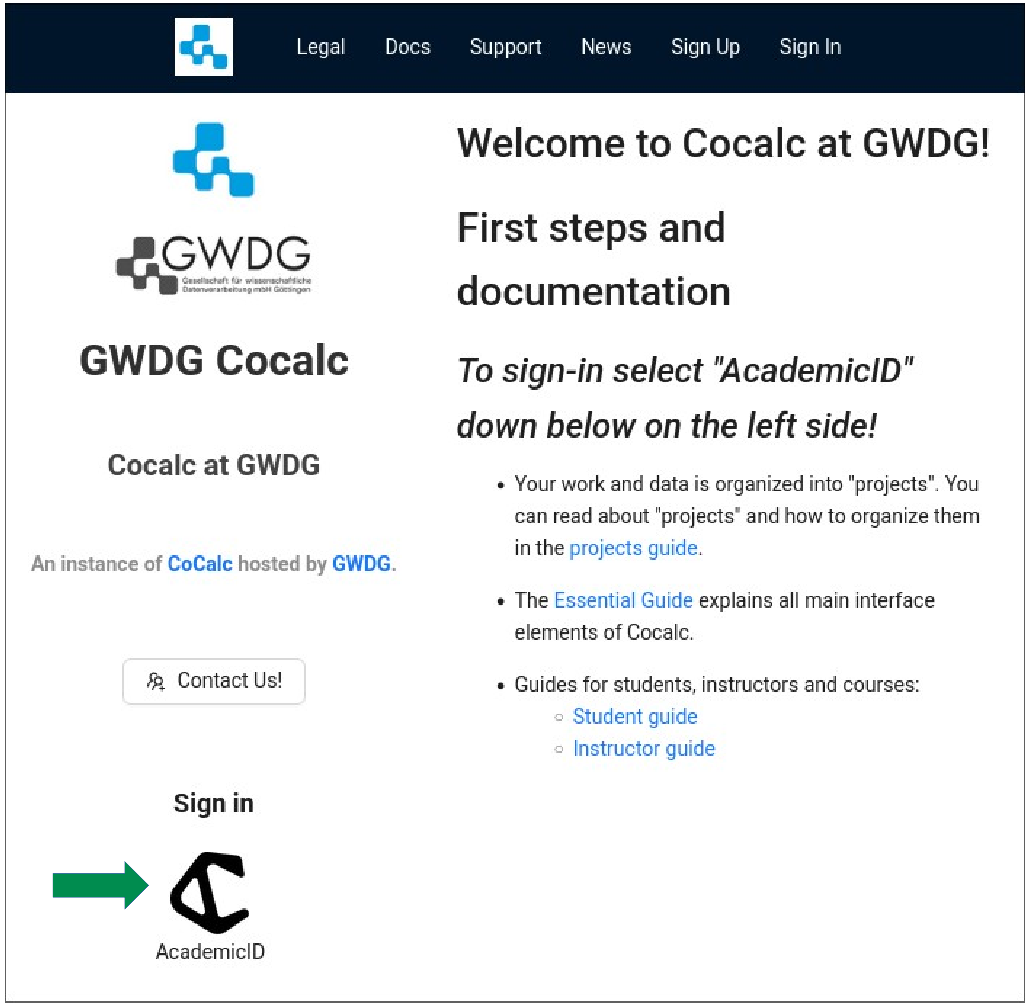
- Exercises are graded semi-automatically. Due to this it is highly important that you follow the required submission format. Otherwise the grading process will fail and you will receive 0 points.

- Exercises will be provided to you via Cocalc https://cocalc.gwdg.de/ with additional instructions provided via GitHub.



Coding Exercise Submission and Grading

- Sign-in using your university credentials.
  1. Click on the "Academic ID" Logo



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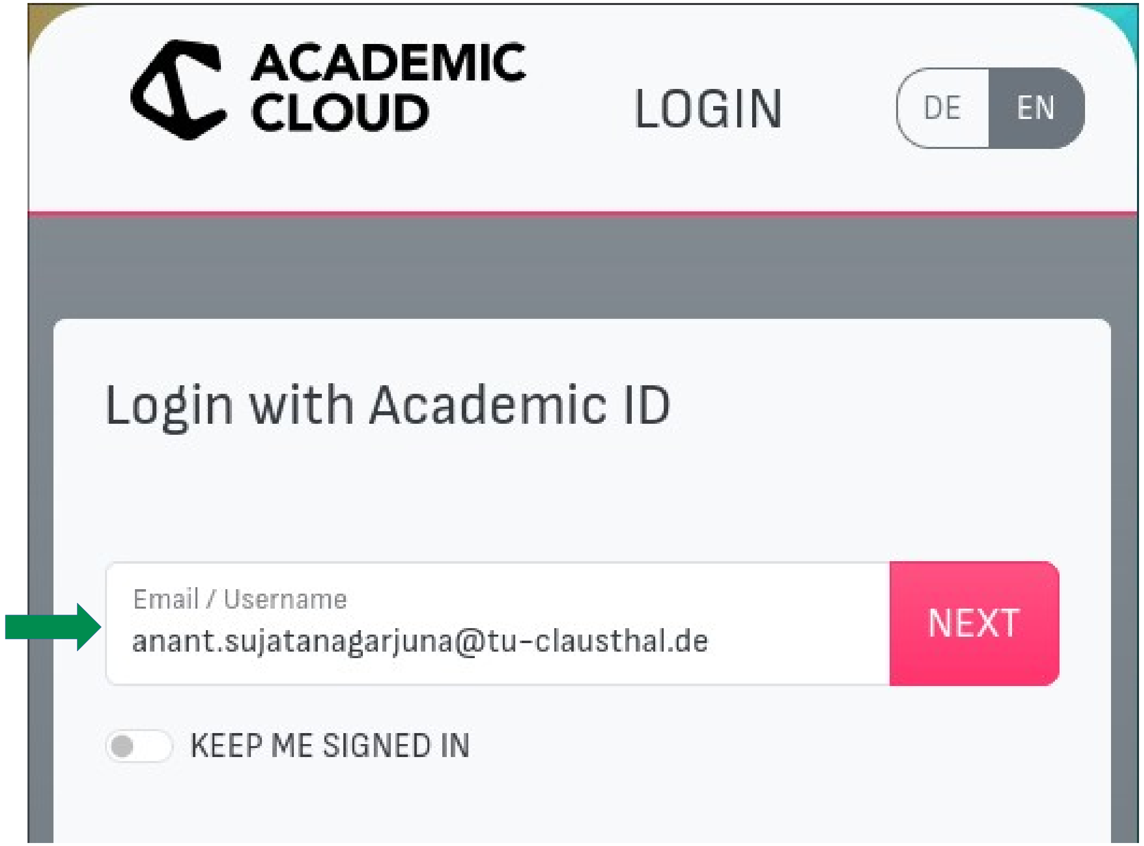

Coding Exercise Submission and Grading

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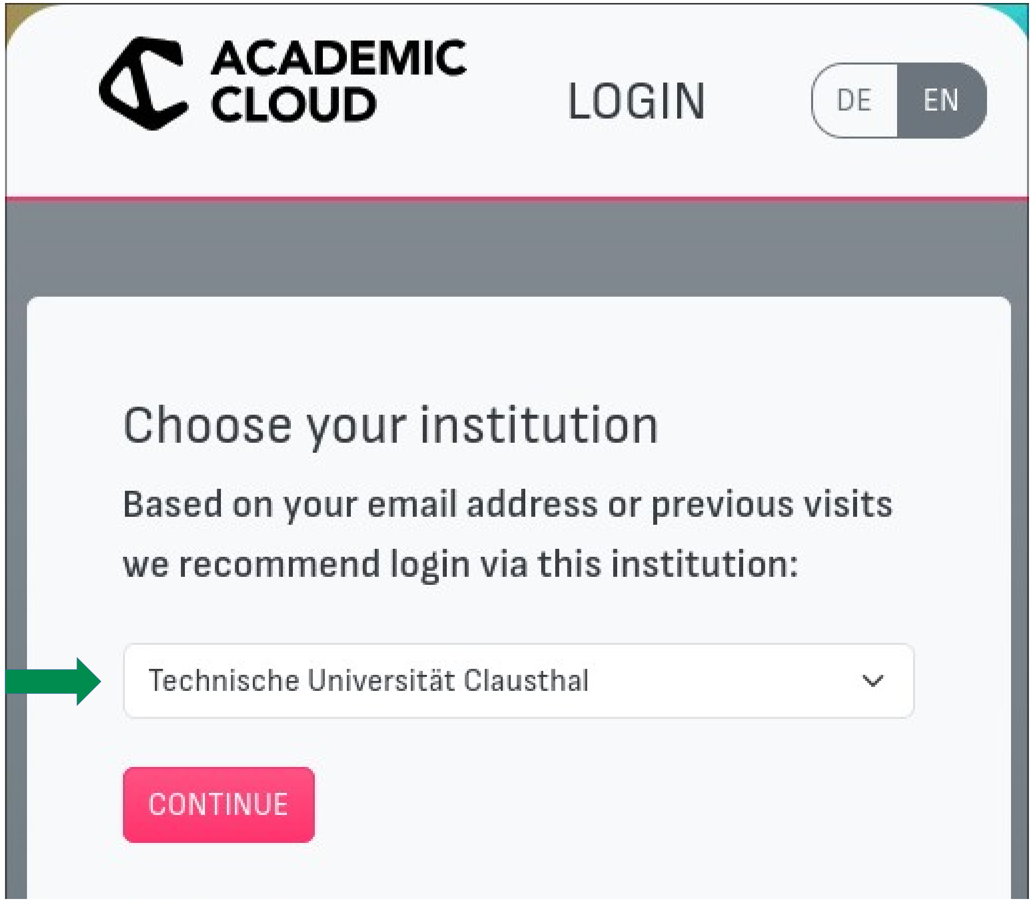
  2. Enter your university email address.
  3. Choose “Technische Universität Clausthal” from the drop-down box.



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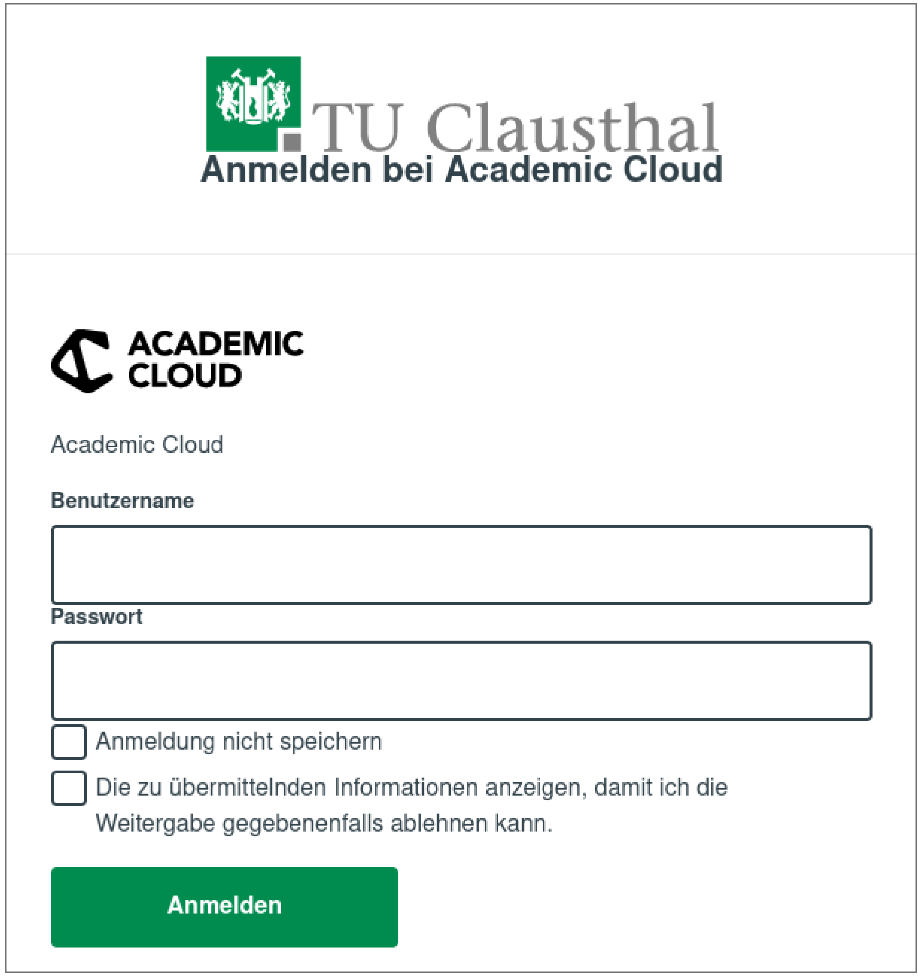
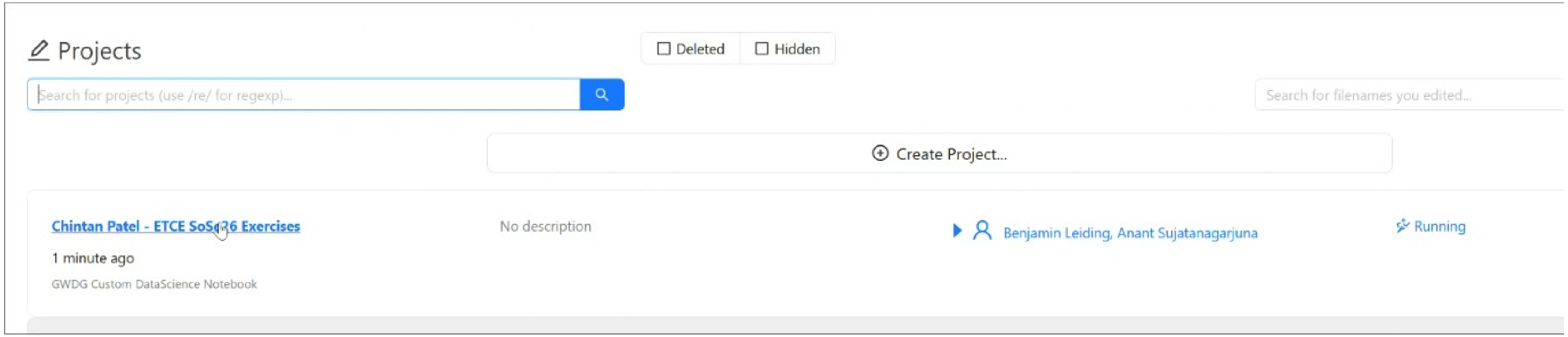
  4. Login with your usual TU-Clausthal credentials.



Coding Exercise Submission and Grading

- Submitting Assignments
  1. Click on “IoT Mining SoSe26”



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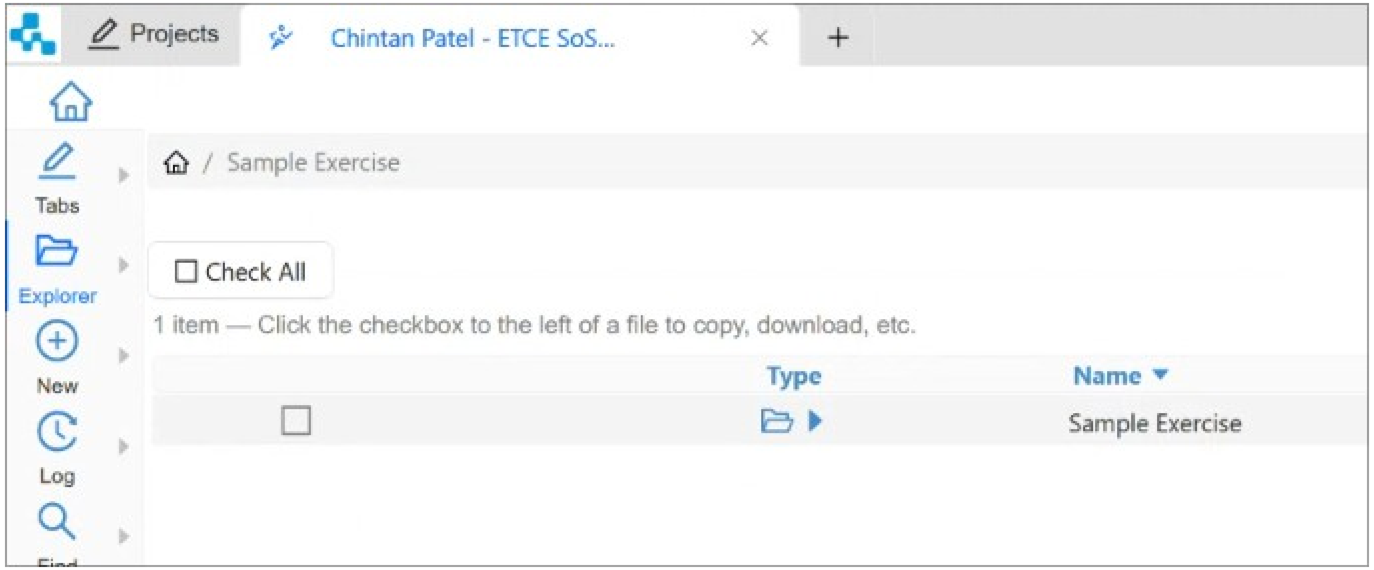
  2. Open the folder of the exercise you want to work on.



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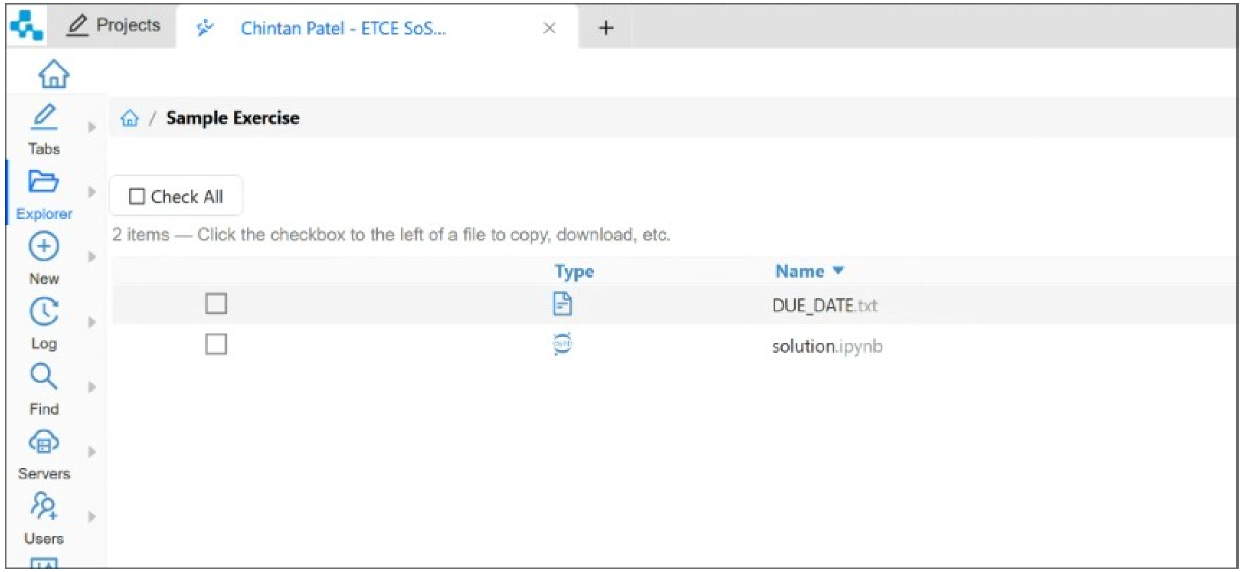

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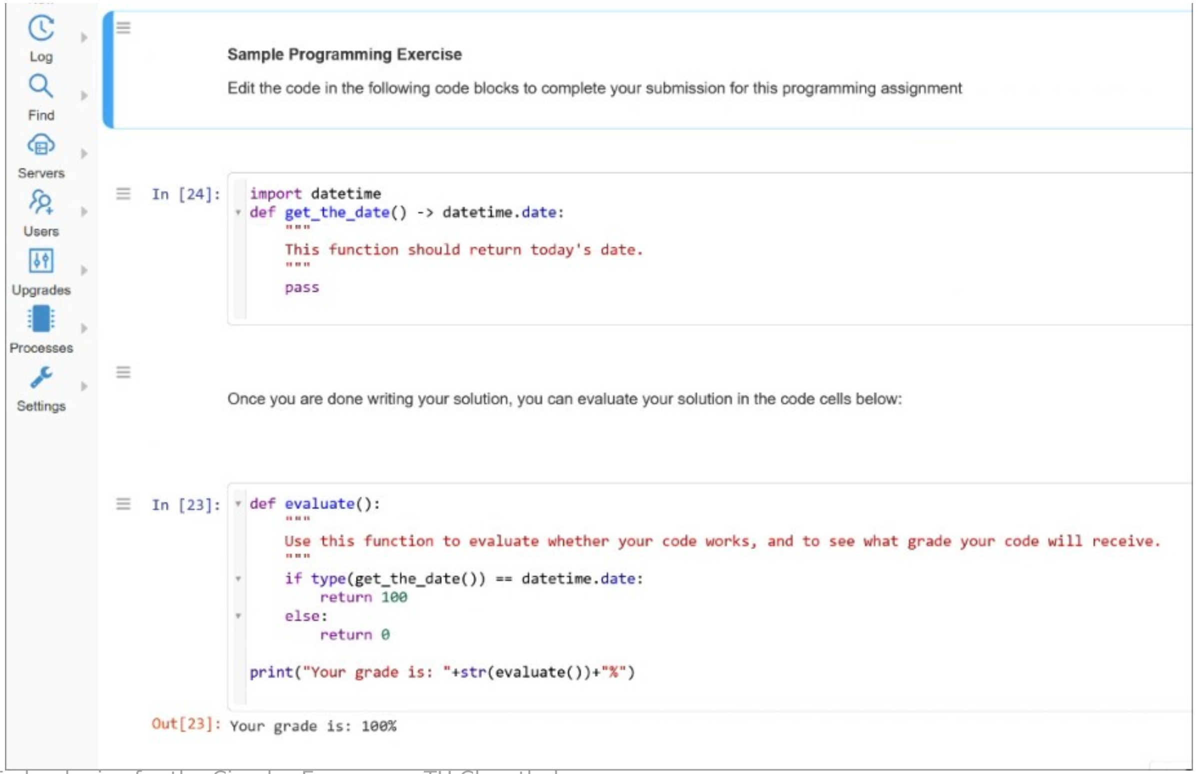

Coding Exercise Submission and Grading

- Exercise Grades
  - We will collect your exercises after the deadline has elapsed.
  - Please ensure to save all changes to the exercise folder BEFORE the deadline.
  - DO NOT submit your assignments via email/other means.

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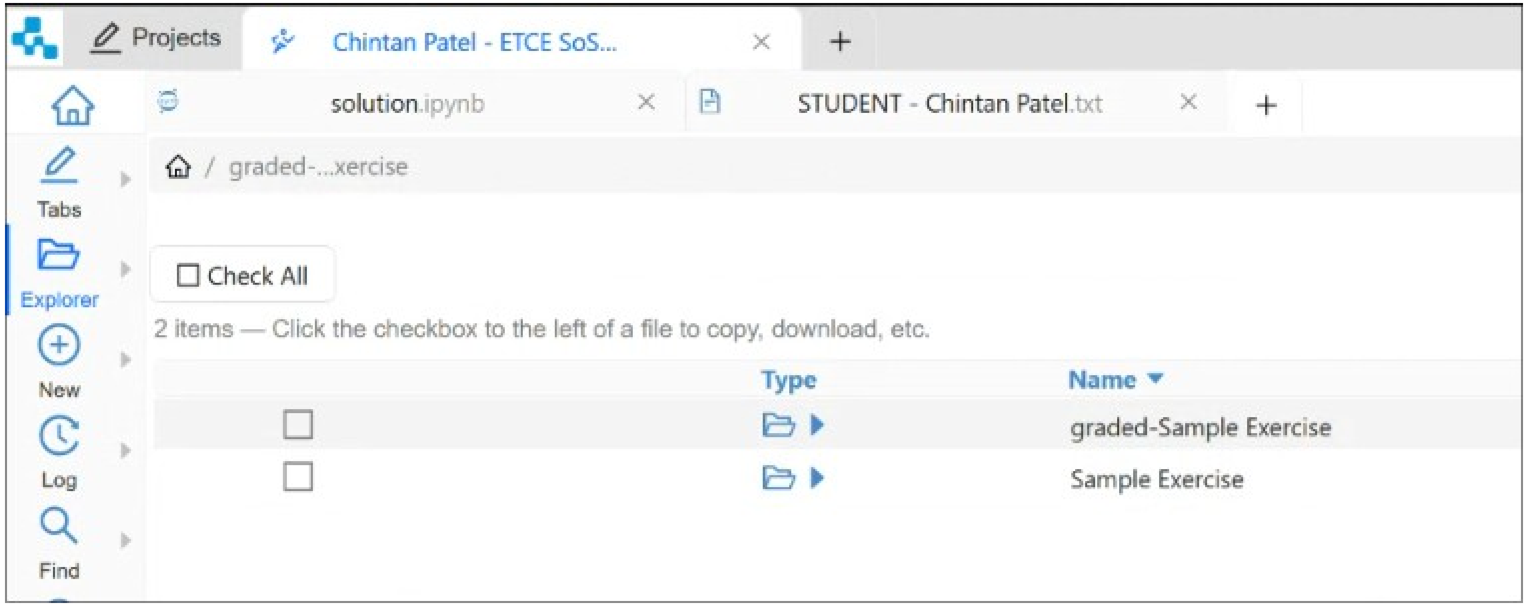

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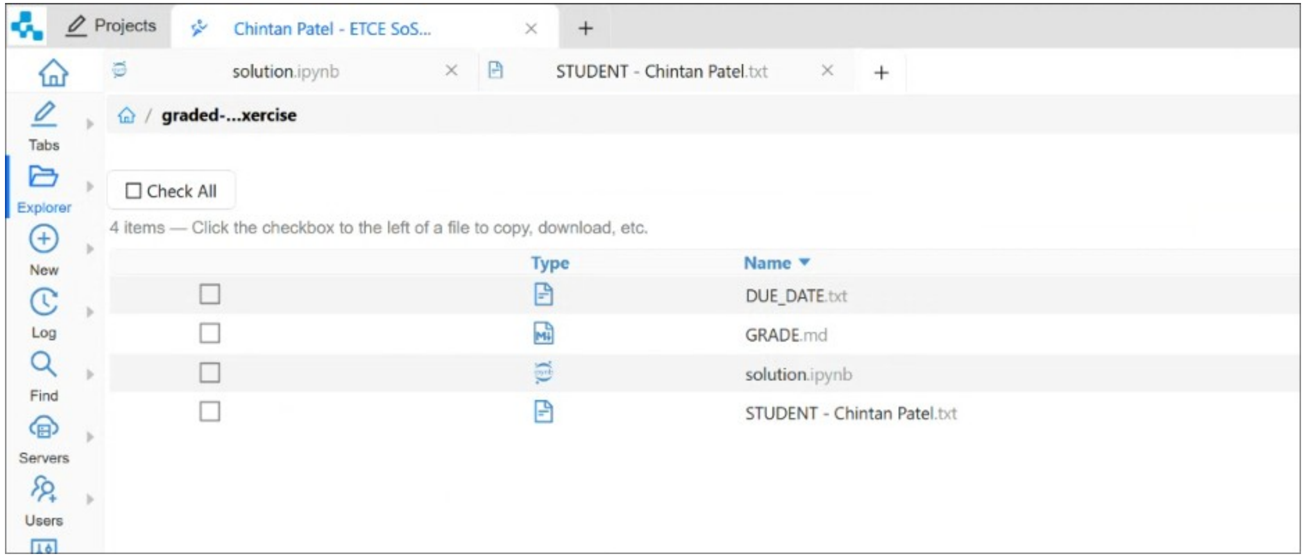

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  - Your grade is provided inside the “GRADE.md” file.
  - Receiving a grade of 0 = FAIL.

Multiple-Choice Exercises

Every student enrolled in this course is advised to take the knowledge quiz in the second week of the course.

- **Goal of the test:**
  - To check the knowledge level of the student that is relevant to this course of study.
- **Preparation:**
  - A review of basic concepts of the Circular Economy is recommended for Week 2.
- **THE TEST IS JUST FOR YOU → WE CANNOT CHECK THE TEST RESULTS**

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Link for all multiple-choice exercises → etce.etce-lab.de

Multiple-Choice

E01: Prior Knowledge

Introduction

This question is the exercise for *"L01 - Introduction"* of the "Emerging Technologies for the Circular Economy" course taught at the TU Clausthal. All the course material is hosted on our GitHub page. This multiple-choice quiz tests your prior knowledge of the topics that we will discuss during the course.

You must answer at least half **the questions** correctly to pass the exercise.

If you have any questions, you can contact us via e-mail: etce-etce@clausthal.de

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Question 01: Block ciphers are used in symmetric cryptography to encrypt and decrypt data. They operate on blocks of a fixed length (usually 64 or 128 bits). To encrypt some data, you could split it into block sized chunks and simply encrypt them separately. What which of the following properties are true?

Choose one or more options then select Submit.

- [ ] This is called ECB mode
- [ ] This is called CBC mode
- [ ] Patterns in the data will show up, which can impact security
- [ ] There is no integrity protection
- [ ] It will be easy to calculate the key

Submit

Multiple-Choice Exercises

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- This is called ECB mode
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- Patterns in the data will show up, which can impact security
- ✓ There is no integrity protection
- It will be easy to calculate the key

Show correct answer

Examination

- Prerequisite for admission to the final exam (all criteria have to be fulfilled):
  - Submit all exercises
  - Pass the practical workshop
- Final exam:
  - **Most likely 12.08.2026 and 13.08.2026**
  - Oral examination (20min)
  - Online

Self-Study Star

Self-Study Star →

- Slides with the self-study star indicate optional/additional study material that is **not** mandatory but could be helpful or interesting

Literature

- This course is not based on a single book and you **do not** need to buy a book to pass the exam.

- Donella H. Meadows, Jorgen Randers, and Dennis L. Meadows. *The Limits to Growth* (1972).
- Donella H. Meadows, Jorgen Randers, and Dennis L. Meadows. *Limits To Growth: The 30-Year Update* (2004).
- Baccini et al. *Metabolism of the Anthroposphere: Analysis, Evaluation, Design* (2012).
- Walter R. Stahel. *The Circular Economy: A User's Guide* (2019).
- W. Brian Arthur. *The Nature of Technology: What It Is and How it Evolves* (2011)
- David Wallace-Wells. *The Uninhabitable Earth, Annotated Edition* (2017).
- (German) Stefan Rahmstorf, Hans Joachim Schellnhuber. *Der Klimawandel* (2019).

Literature

- Perry Lea. Internet of Things for Architects: Architecting IoT solutions by implementing sensors, communication infrastructure, edge computing, analytics, and security (2018).
- M.A. Khan, M.T. Quasim, F. Algarni, A. Alharthi. *Decentralised Internet of Things* (2020).
- Dimitrios Serpanos und Marilyn Claire Wolf. *Internet-of-Things (IoT) Systems Architectures, Algorithms, Methodologies* (2018).

Questions?